

Alessandro Metelli

Software Engineer | Backend, Developer Infrastructure, AI Tooling

Remote (GMT-Compatible)

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Professional Summary

Software Engineer with 5 years of experience building backend systems, developer tooling, blockchain infrastructure, and AI-assisted engineering workflows. Strong experience with Rust, TypeScript, Python, Go, and distributed systems. Contributed to open-source developer platforms, blockchain protocols, contract verification tooling, and developer productivity systems. Comfortable owning projects from architecture and implementation through deployment, debugging, and production support.

Selected Achievements

- Built and maintained infrastructure and developer tooling used across the Starknet ecosystem.
 - Contributed contract verification functionality to Starknet Foundry.
 - Developed cross-chain infrastructure integrating Starknet, LayerZero, Hyperlane, and AggLayer.
 - Built Open Maintainer, an open-source platform for AI-assisted repository maintenance, PR review, and issue triage.
 - Designed and developed NoGame, a fully on-chain MMO, as a solo engineer.
 - Built production tooling in Rust, TypeScript, Python, Go, Solidity, and Cairo.
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Experience

Nethermind — Software Engineer

Apr 2024 – Jun 2026

Worked on blockchain infrastructure, backend services, developer tooling, smart contracts, and AI-assisted engineering workflows.

Key Contributions

- Built backend systems and developer tooling using Rust, TypeScript, Python, Cairo, and Solidity.
- Developed AggSandbox, a Rust CLI for local multichain infrastructure, transaction execution, and bridge debugging.
- Maintained Voyager Verifier, improving contract verification workflows, compiler compatibility, metadata validation, and developer diagnostics.
- Contributed `sncast verify` functionality to Starknet Foundry.
- Built integrations across Starknet, LayerZero, Hyperlane, AggLayer, and EVM ecosystems.
- Developed components for ZK infrastructure, including Groth16 verification and Ethereum-to-Starknet data proving systems.
- Built internal AI-assisted engineering workflows focused on automation, validation, and developer productivity.

NoGame — Lead Developer

Jan 2022 – Present

Founder and lead engineer of a fully on-chain MMO.

- Designed and implemented backend services, frontend applications, APIs, smart contracts, game systems, and infrastructure.

- Built React and TypeScript user interfaces, analytics systems, and blockchain integrations.
 - Owned architecture, product design, implementation, deployment, and ongoing operation.
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Open Source Contributions

Starknet Foundry

Contributed contract verification tooling and developer workflow improvements.

Apibara

Contributed to Rust-based blockchain indexing infrastructure and event-processing systems.

Dojo Engine

Built tooling and integrations for Starknet application development.

Voyager Verifier

Developed and maintained Starknet smart contract verification infrastructure.

Selected Projects

Open Maintainer

Open-source platform for repository audits, AI-ready repository context, pull request review, and issue triage.

Tech: TypeScript, Bun, React, Fastify, PostgreSQL, Redis

Horizon

Yield-tokenization protocol on Starknet implementing PT/YT mechanics inspired by Pendle.

Tech: Cairo, TypeScript, React

weld-cli

Human-in-the-loop AI coding harness designed around traceability, validation, and controlled execution.

Tech: Python

AggSandbox

Rust CLI for local multichain infrastructure, bridge testing, and transaction inspection.

Tech: Rust

Technical Skills

Languages: Rust, TypeScript, JavaScript, Python, Go, Solidity, Cairo, Noir

Backend: Node.js, REST APIs, PostgreSQL, Redis, Event-Driven Systems, Distributed Systems

Frontend: React, TypeScript

Infrastructure: Docker, Linux, GitHub Actions, CI/CD, AWS

AI Tooling: Agent Workflows, Tool Orchestration, Browser Automation, Human-in-the-Loop Systems

Blockchain: Starknet, Ethereum, Aztec, LayerZero, Hyperlane, AggLayer, Smart Contracts, DeFi, ZK Systems

Education

BSc Data Science and Computing

Birkbeck, University of London